BASIC EQUATIONS

Polytropic Processes ($PV^n = \text{const}$)

for $n = 1$

$$W = P_1 V_1 \log \frac{V_2}{V_1}$$

for $n \neq 1$

$$W = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

$$\gamma = \frac{c_p}{c_v} \qquad c_p = c_v + R_{sp}$$

Two-phase Mixtures

the following equations are valid, where y can be v, u, h, or s

$$y = \frac{Y}{m} \qquad y = y_f + x(y_g - y_f)$$

$$x = \frac{m_g}{m_f + m_g} = \frac{y - y_f}{y_g - y_f}$$

Non-ideal Liquids

the following equations are valid, where y can be v, u, or s

$$y(T, P) \approx y_f(T)$$

for h

$$h(T, P) \approx h_f(T) + v_f(T)\big(P - P_{sat}(T)\big)$$

Interpolation

$$x_1 < x < x_2 \qquad \frac{y(x) - y(x_1)}{y(x_2) - y(x_1)} = \frac{x - x_1}{x_2 - x_1}$$

HEAT TRANSFER

$$\dot{Q} = \frac{1}{R}\Delta T$$

Conduction

$$R = \frac{\Delta x}{kA}$$

Convection

$$R = \frac{1}{hA}$$

Radiation

$$1/R = \varepsilon\sigma A\big(T_{\text{surf}}^2 + T_{\text{amb}}^2\big)(T_{\text{surf}} + T_{\text{amb}})$$

$$\dot{Q}_{\text{in}} = \varepsilon\vec{A} \cdot \vec{S}$$

$$\dot{Q}_{\text{in}} = \varepsilon\sigma A T_{\text{amb}}^4$$

$$\dot{Q}_{\text{out}} = \varepsilon\sigma A T_{\text{surf}}^4$$

FLUID FLOWS

Viscosity

$$\tau = \frac{F_t}{A} = \mu\frac{d\mathcal{V}}{dy}$$

$$\mathcal{V}_z(r) = 2\mathcal{V}_m\left(1 - \left(\frac{r}{R}\right)^2\right) \qquad \mathcal{V}_m = \frac{R^2}{8\mu}\left(\frac{\Delta P}{L} - \rho g \sin\theta\right)$$

Turbulent Flow ($Re > 4000$; $Re > 2100$ transitional)

$$f \equiv \frac{\Delta P^*}{L^*} = g(Re)$$

$$Re = \frac{\rho L_{char}\mathcal{V}}{\mu}$$

Conservation and Balance Equations

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e$$

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i\left(h_i + \frac{1}{2}\mathcal{V}_i^2 + gz_i\right) - \dot{m}_e\left(h_e + \frac{1}{2}\mathcal{V}_e^2 + gz_e\right)$$

$$\frac{d\vec{p}_{cv}}{dt} = \vec{F}_{net} + \dot{m}_i\vec{\mathcal{V}}_i - \dot{m}_e\vec{\mathcal{V}}_e$$

$$\frac{dS_{cv}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen} + \dot{m}_i s_i - \dot{m}_e s_e$$

Continuity equation $\rho_i A_i \mathcal{V}_i = \rho_e A_e \mathcal{V}_e$

Bernoulli equation $P_i + \frac{1}{2}\rho\mathcal{V}_i^2 + \rho g z_i = P_e + \frac{1}{2}\rho\mathcal{V}_e^2 + \rho g z_e$

Open-System Devices

Turbines	$\dot{W}_{cv} = \dot{m}(h_i - h_e)$	$\epsilon_T = \frac{\dot{W}_{act}}{\dot{W}_{ideal}} = \frac{h_1 - h_2}{h_1 - h_{2s}}$
Compressors/Pumps	$-\dot{W}_{cv} = \dot{m}(h_e - h_i)$	$\epsilon_{C/P} = \frac{\dot{W}_{ideal}}{\dot{W}_{act}} = \frac{h_{2s} - h_1}{h_2 - h_1}$
Ideal Pumps	$-\dot{W}_{cv} = \dot{m}v(P_e - P_i)$	$\epsilon_P = \frac{\dot{W}_{ideal}}{\dot{W}_{act}} = \frac{v(P_2 - P_1)}{h_2 - h_1}$
Nozzles, Diffusers	$h_i + \frac{1}{2}\mathcal{V}_i^2 = h_e + \frac{1}{2}\mathcal{V}_e^2$	
Throttling Values	$h_i = h_e$	
Mixing Chambers	$\sum \dot{m}_i h_i + \dot{Q}_{cv} - \dot{W}_{cv} = \dot{m}_e h_e$	
Heat Exchangers	$\dot{m}_H(h_1 - h_2) = \dot{m}_L(h_4 - h_3)$	
	$\dot{Q}_H = \dot{m}_H(h_1 - h_2)$	
	$\dot{Q}_L = \dot{m}_L(h_4 - h_3)$	

Ideal Vapor-Compression Cycle

In this space, I use a T-s graph of an ideal vapor-compression cycle from *Introduction to Thermal and Fluids Engineering*, by Deborah A. Kaminski and Michael K. Jensen, Wiley, 2017, Fig 8-2, pg 319

Ideal Rankine Cycle

In this space, I use a T-s graph of an ideal Rankine cycle from *Introduction to Thermal and Fluids Engineering*, by Deborah A. Kaminski and Michael K. Jensen, Wiley, 2017, Fig 8-8, pg 332

$$\epsilon_{\text{boiler}} = \frac{\dot{Q}_H}{\dot{m}_{\text{fuel}}\, HV}$$